Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence IT3021

Hotel Booking Analysis Data Warehouse SSAS & BI Submission Report

Assignment 2

2026

Student ID : IT23194908
Student Name : R A S R Harischandra
Batch / Group : Y3.S2.DS.WE.02.02

Table of Contents

1.Data source

1.1 Database Overview

The data source is the **Hotel_DW** SQL Server database designed and populated in Assignment 1. It implements a **Star Schema** centred on the **Fact_Reviews** table, which stores 49,999 hotel review records spanning the years 2020–2025. Three-dimension tables and one central fact table make up the warehouse.

Dataset selected – Hotel Booking Analytics

Schema Summary

Table	Type	Key Column	Description
Fact_Reviews	Fact	review_id (PK)	49,999 review transactions — 7 score measures + 3 accumulating columns
Dim_User	Dimension (SCD2)	user_key (PK)	2,001 users — gender, country, age group, traveller type (SCD Type 2)
Dim_Hotel	Dimension	hotel_key (PK)	25 hotels across 25 cities in 25 countries
Dim_Date	Dimension	date_key (PK)	Date dimension — year, month, day (2020–2025)

1.2 Dimension Tables

Dim_Hotel

Column	Type	Description
hotel_key	INT (PK)	Surrogate primary key
hotel_id	INT	Natural key
hotel_name	VARCHAR(100)	Hotel name (e.g. Kyo-to Grand)
city	VARCHAR(50)	City — used in Geography hierarchy
country	VARCHAR(50)	Country — top of Geography hierarchy
star_rating	INT	Hotel star rating (all 5-star in this dataset)

Dim_User_(SCD Type 2)

Column	Type	Description
user_key	INT (PK)	Surrogate primary key (auto-increment)
user_id	INT	Natural key from source
gender	VARCHAR(10)	Male / Female
country	VARCHAR(50)	User's home country
age_group	VARCHAR(20)	18–24 / 25–34 / 35–44 / 45–54 / 55+
traveller_type	VARCHAR(50)	Solo / Couple / Family / Business / Other
start_date	DATE	SCD2: date this version became active
end_date	DATE	SCD2: date superseded (NULL = current)
is_current	BIT	1 = current row, 0 = historical

Dim_Date

Column	Type	Description
date_key	INT (PK)	Date key in YYYYMMDD format
full_date	DATE	Full calendar date
year	INT	Calendar year (2020–2025)
month	INT	Month number (1–12)
day	INT	Day of month (1–31)

Dim_Location

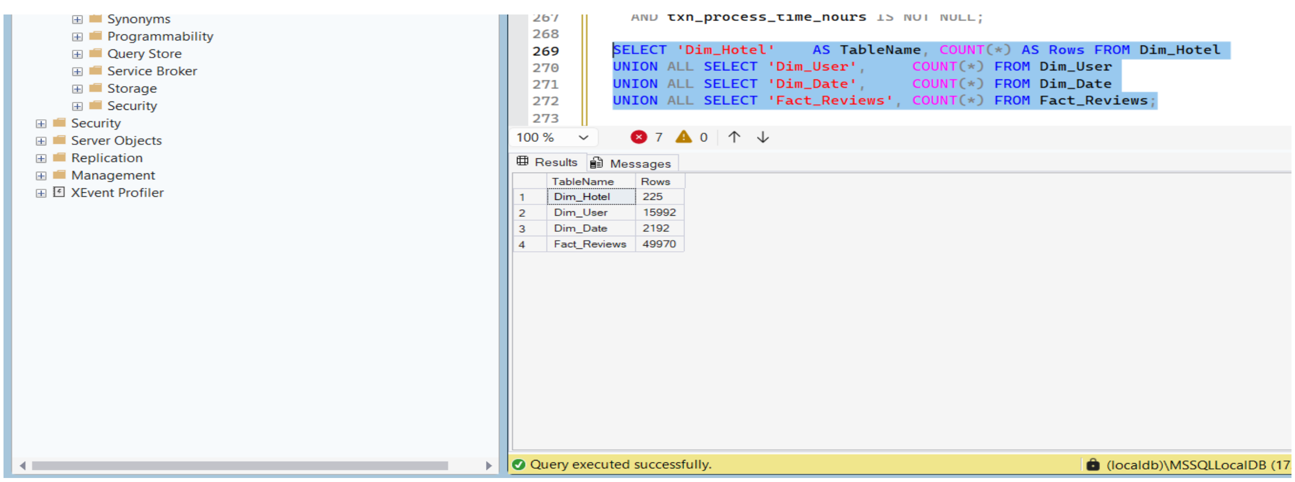
Column	Type	Description
location_id	INT (PK)	Unique identifier for each location
city	VARCHAR	Name of the city where the hotel is located
country	VARCHAR	Country of the hotel
continent	VARCHAR	Continent of the location (e.g., Asia, Europe)
region	VARCHAR	Geographic region (e.g., South Asia, Western Europe)

1.3 Fact Table — Fact_Reviews

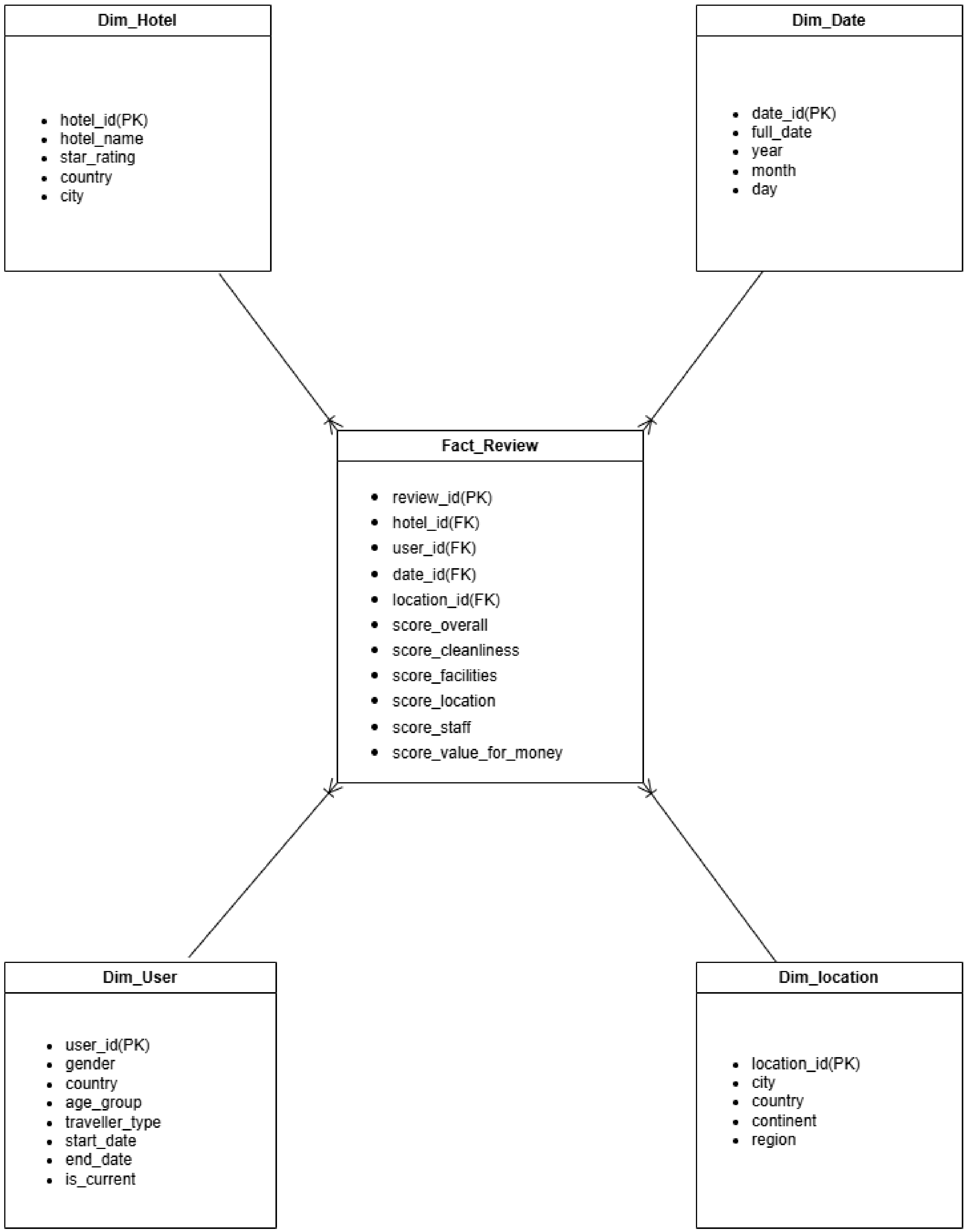
The **Fact_Reviews** table is the centre of the star schema. Each row represents one hotel review event with the following structure.

Column	Type	Description
review_id	INT (PK)	Natural key — unique review identifier
user_key	INT (FK)	Surrogate key → Dim_User
hotel_key	INT (FK)	Surrogate key → Dim_Hotel
date_key	INT (FK)	Surrogate key → Dim_Date
score_overall	FLOAT	Overall review score (1–10) — MEASURE
score_cleanliness	FLOAT	Cleanliness score — MEASURE
score_comfort	FLOAT	Comfort score — MEASURE
score_facilities	FLOAT	Facilities score — MEASURE
score_location	FLOAT	Location score — MEASURE
score_staff	FLOAT	Staff score — MEASURE
score_value_for_money	FLOAT	Value for money score — MEASURE
accm_txn_create_time	DATETIME	Accumulating: record creation timestamp
accm_txn_complete_time	DATETIME	Accumulating: transaction completion timestamp
txn_process_time_hours	INT	Accumulating: hours between create and complete

1.4 Row Count Validation



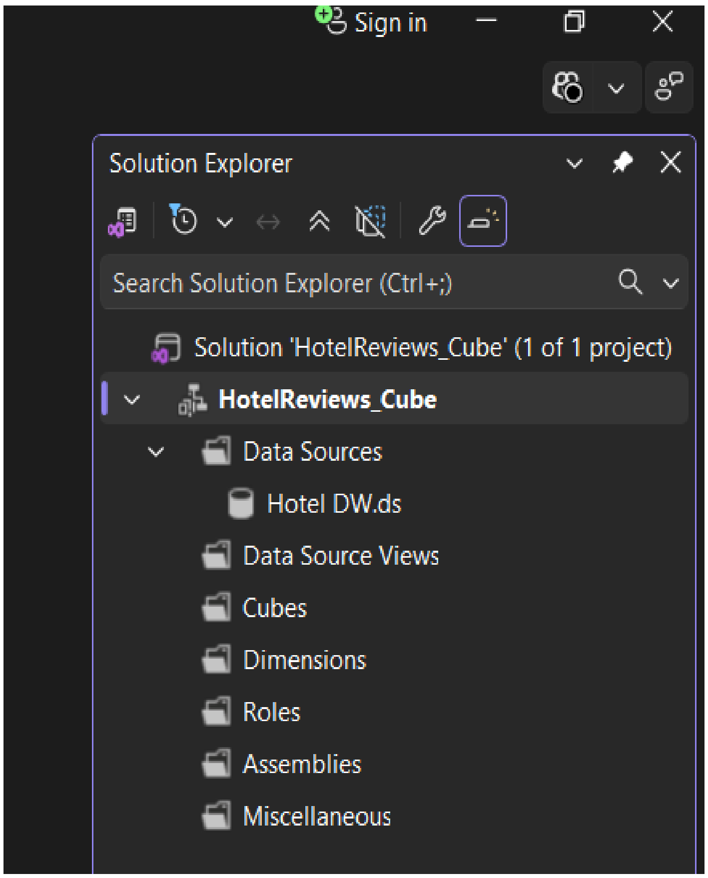
1.5 Star Schema Diagram



2. SSAS Cube Implementation

An SSAS Multidimensional cube named HotelReviews_Cube was built in SQL Server Data Tools (SSDT / Visual Studio) using Hotel_DW as the data source. The cube includes all 7 score measures from Fact_Reviews, all 3-dimension tables, two hierarchies, and one calculated measure for average score.

2.1 Creating the SSAS Project

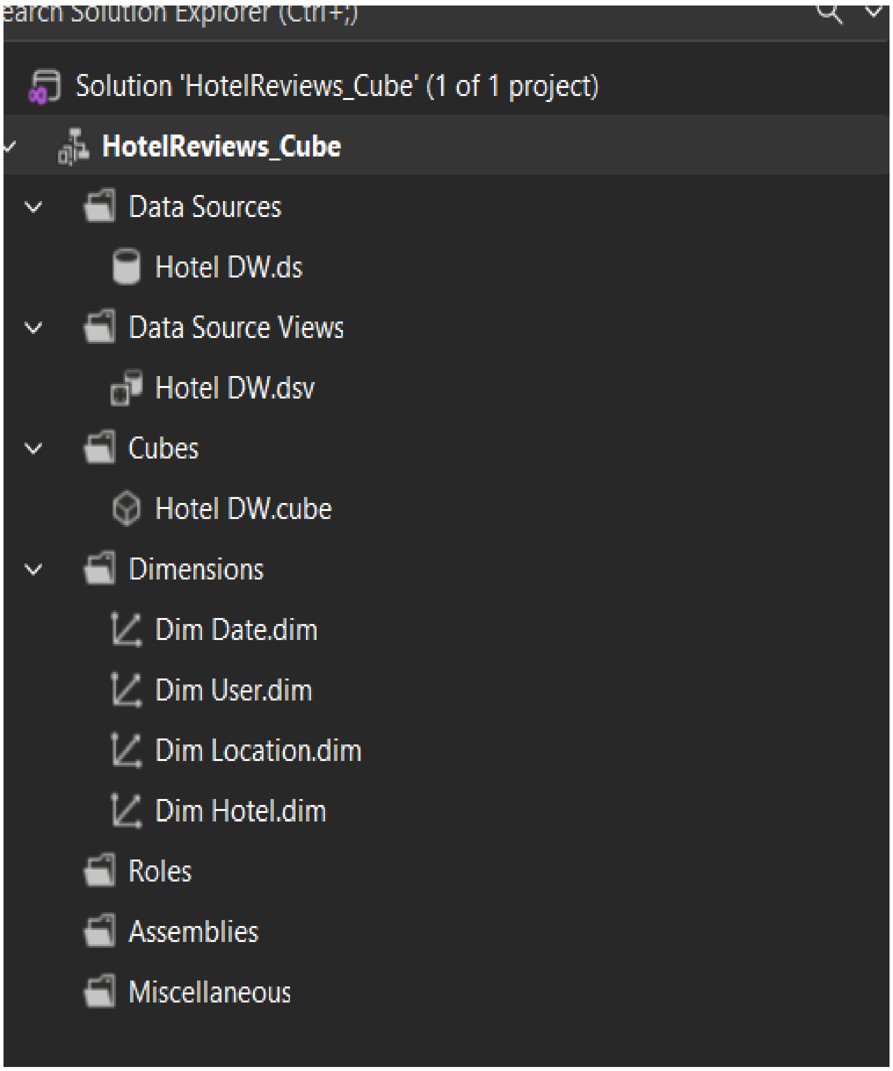


A new Analysis Services Multidimensional and Data Mining Project was created in Visual Studio. The project was named “HotelReviews_Cube” to represent the OLAP cube for hotel review analysis. This project environment is used to design dimensions, cubes, and perform analytical processing.

2.2 Create the Data Source & Data Source View (DSV)

A Data Source connection was established to the Hotel_DW database using Native OLE DB and SQL Server. The connection was successfully tested to ensure proper communication between SSAS and the relational database. The service account impersonation option was selected to allow secure data access.

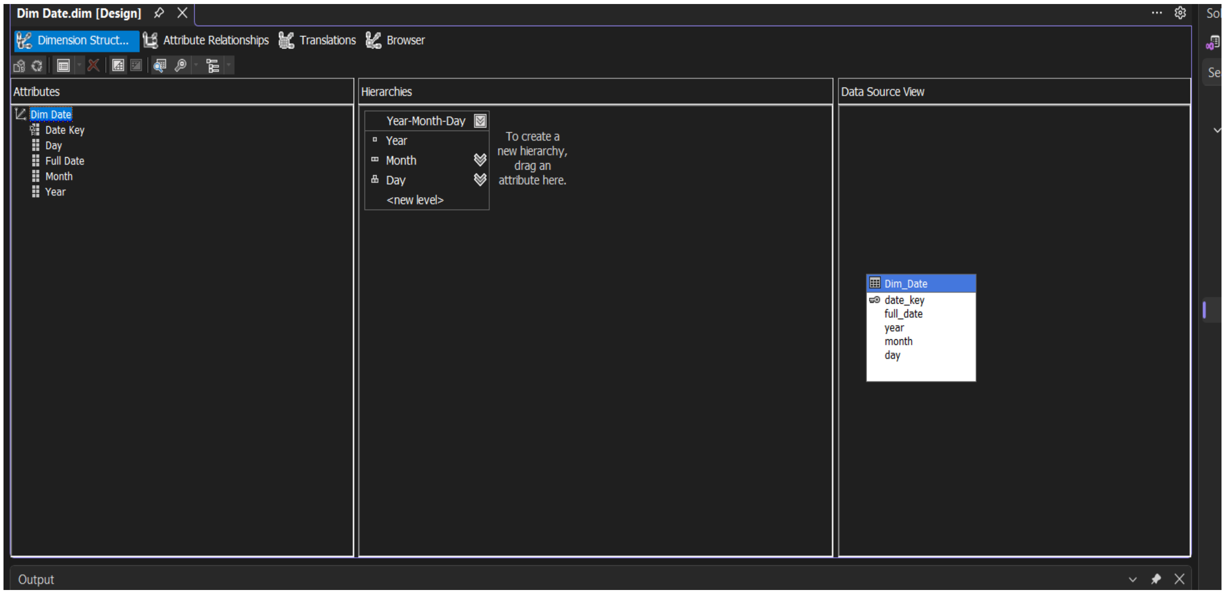
A Data Source View (DSV) was created by selecting Fact_Reviews and all dimension tables. The DSV visually represents the schema and relationships between tables. Foreign key relationships were verified to ensure proper linking between fact and dimension tables.

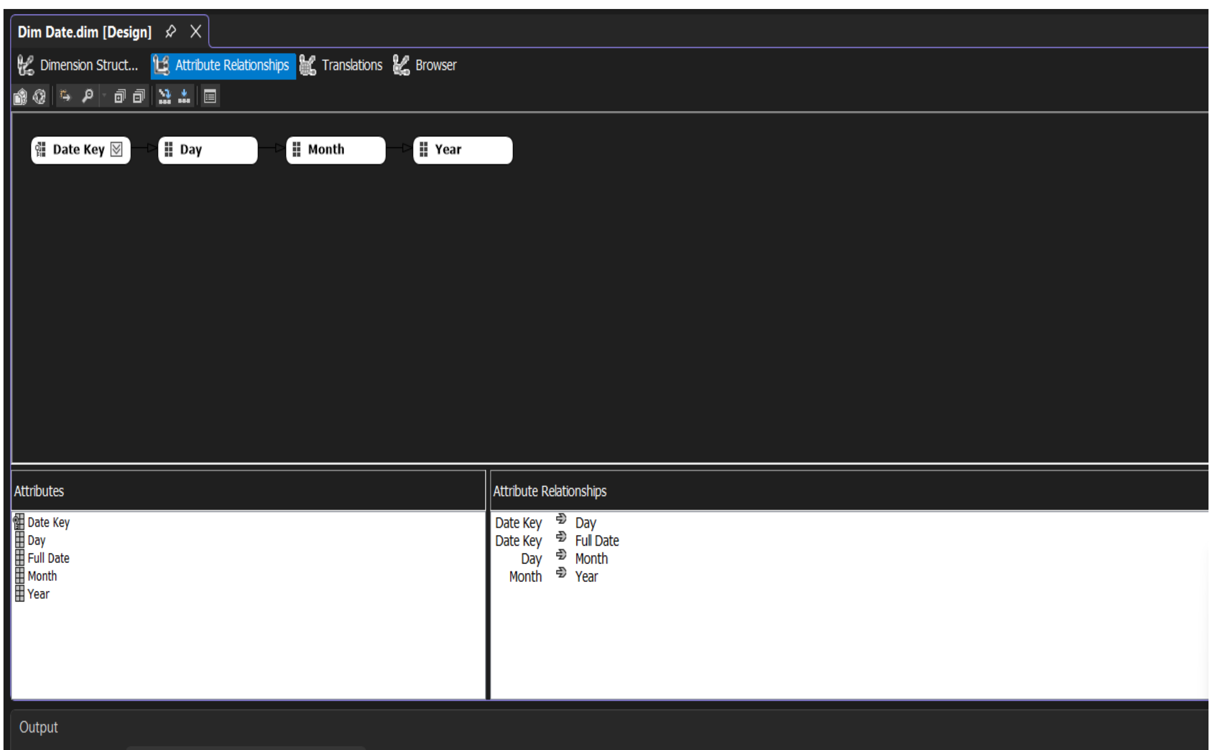


2.3 Creating Dimensions

Dim_Date Dimension

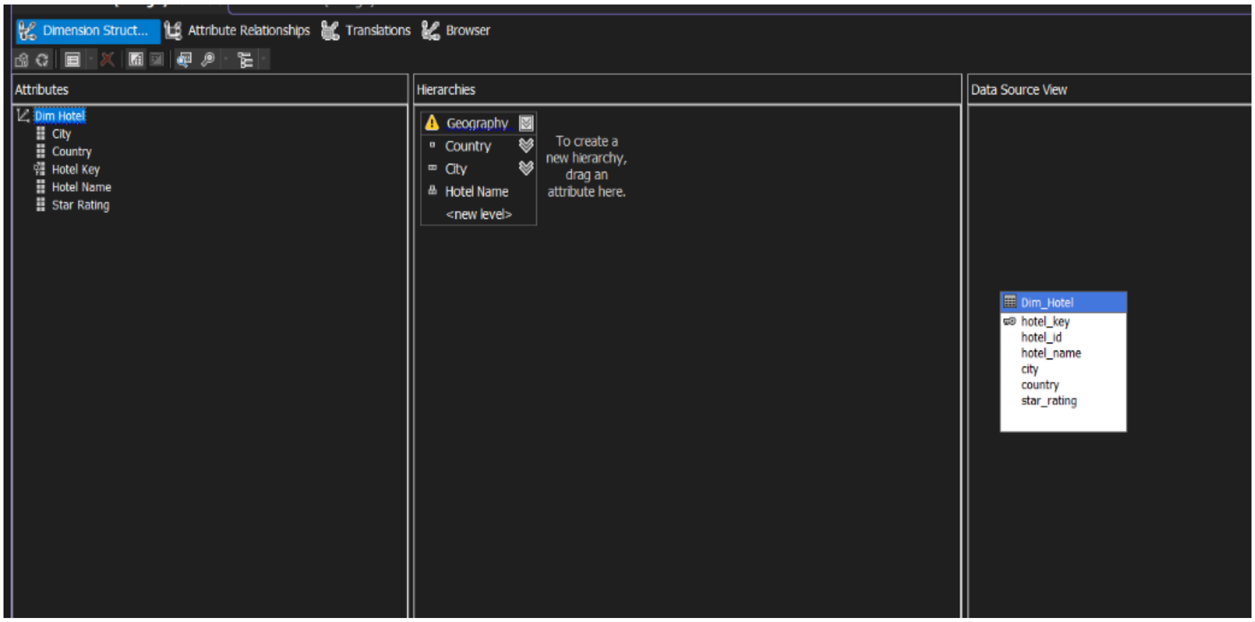
The Dim_Date table was used to create the Date dimension with date_key as the primary key. Attributes such as year, month, and day were included to support time-based analysis. A hierarchy (Year → Month → Day) was created to enable drill-down operations in reports.

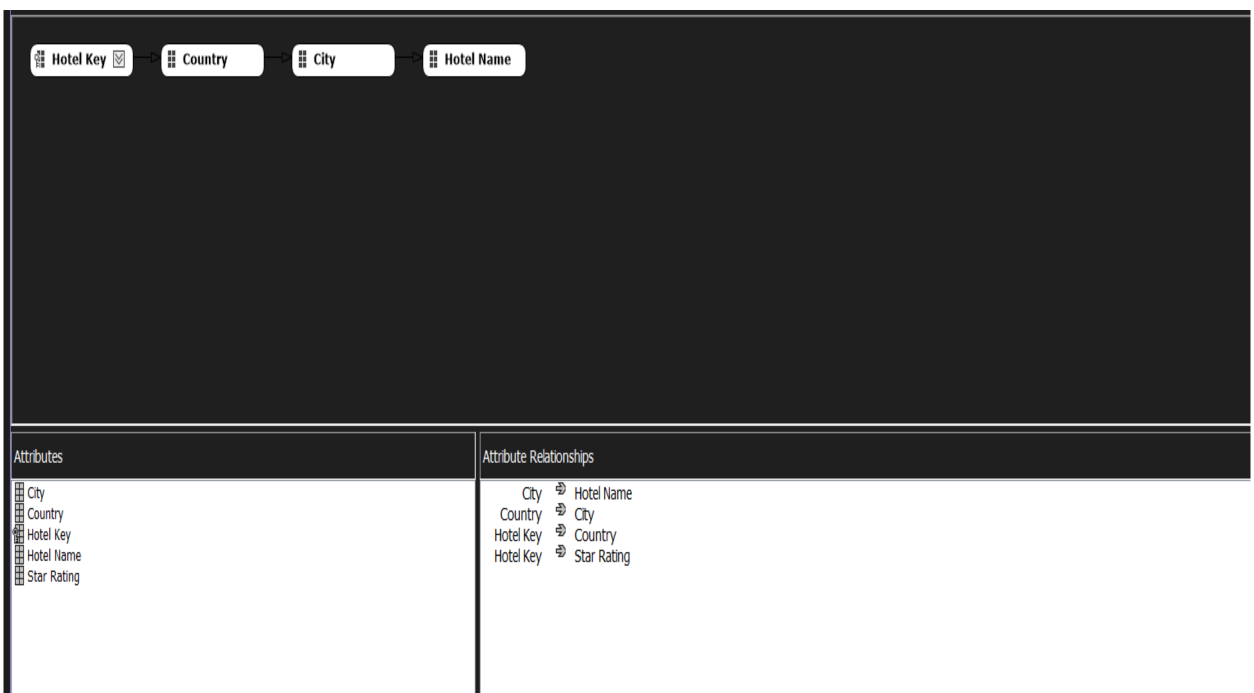




Dim_Hotel Dimension

The Dim_Hotel table was used to create a dimension containing hotel-related information. Attributes such as hotel_name, city, country, and star_rating were selected. A Geography hierarchy (Country → City → Hotel Name) was created to support location-based analysis.

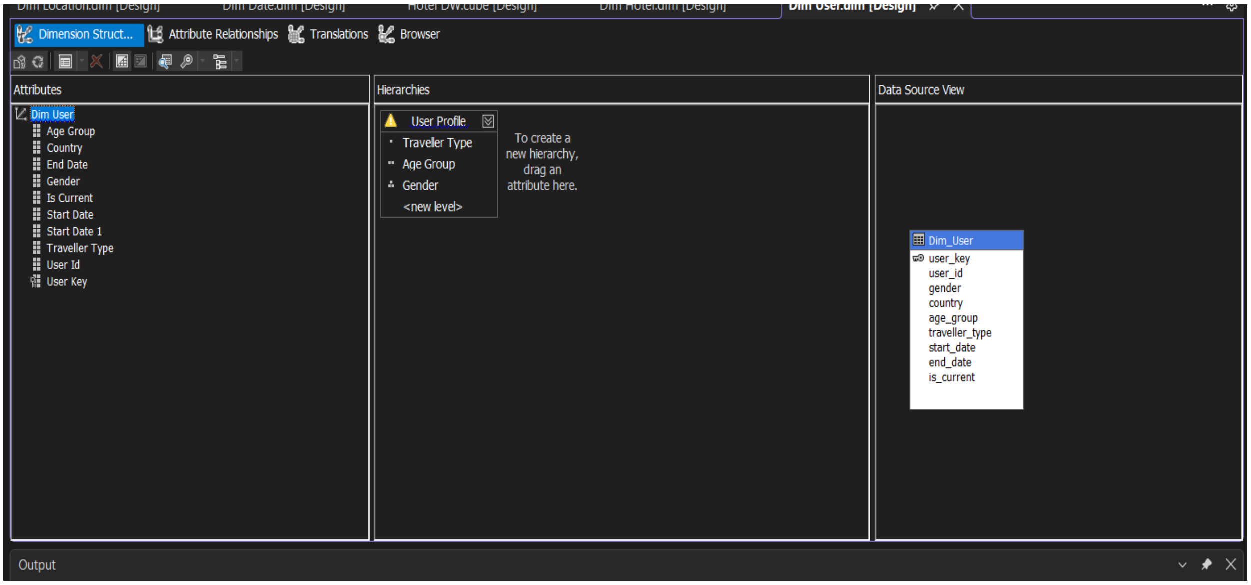


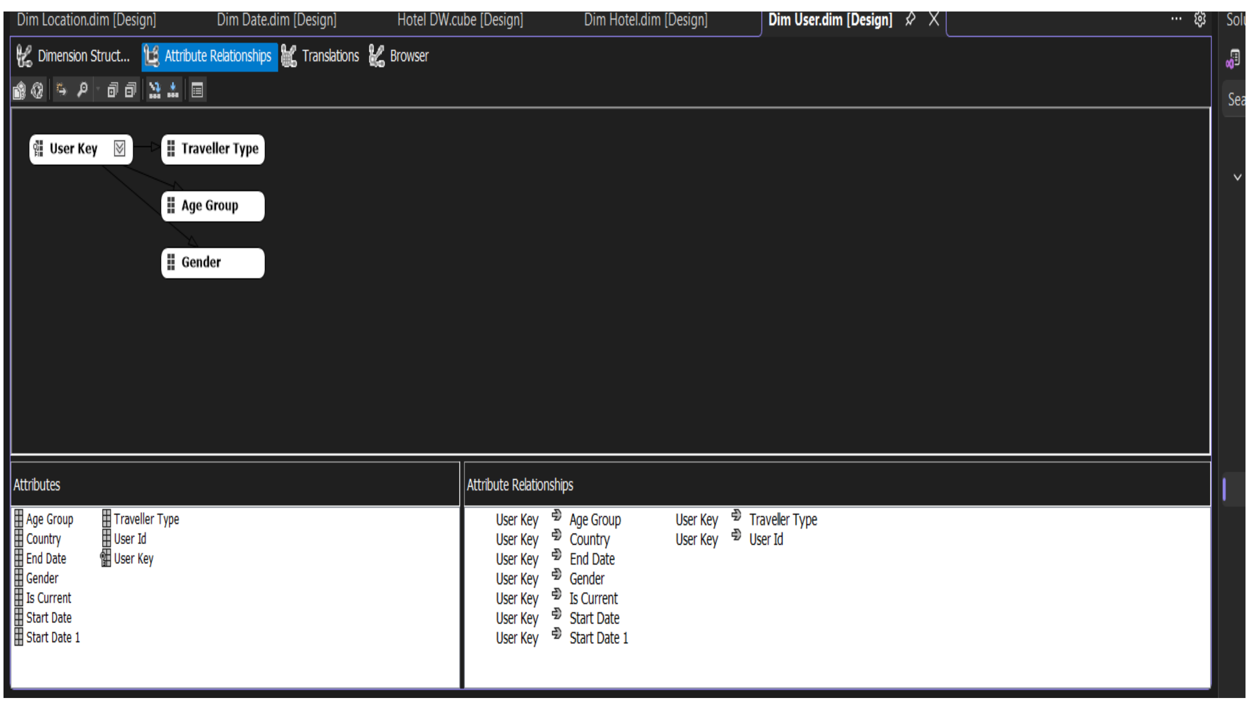


Dim_User Dimension

Dim_User Dimension

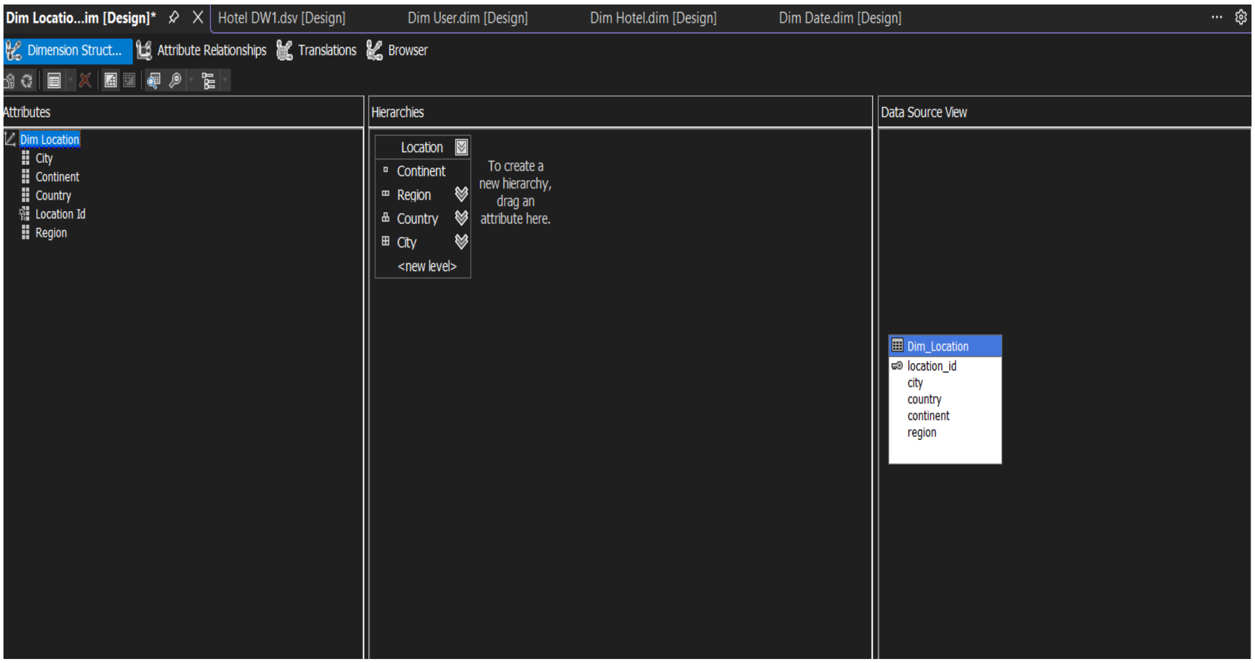
The Dim_User table was used to create a user dimension including demographic attributes such as gender, country, age_group, and traveller_type. The is_current attribute was included to manage slowly changing dimensions (SCD Type 2), ensuring only current records are used in analysis.





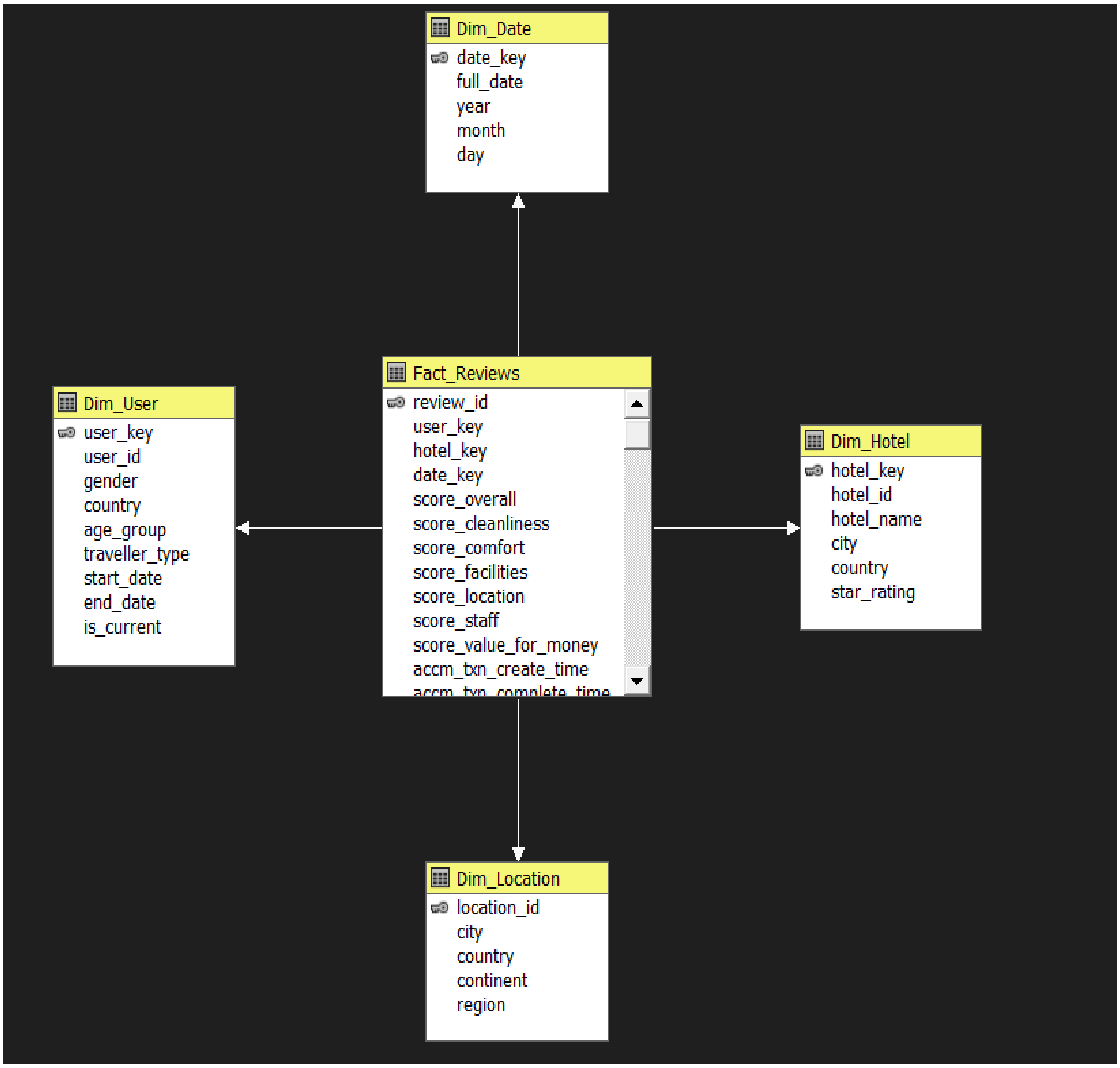
Dim_Location Dimension

The Dim_Location table was used to create a location dimension containing geographical attributes such as city, country, continent, and region. The location_id column was selected as the primary key. This dimension supports advanced geographical analysis and enables users to analyze hotel reviews based on different regions.



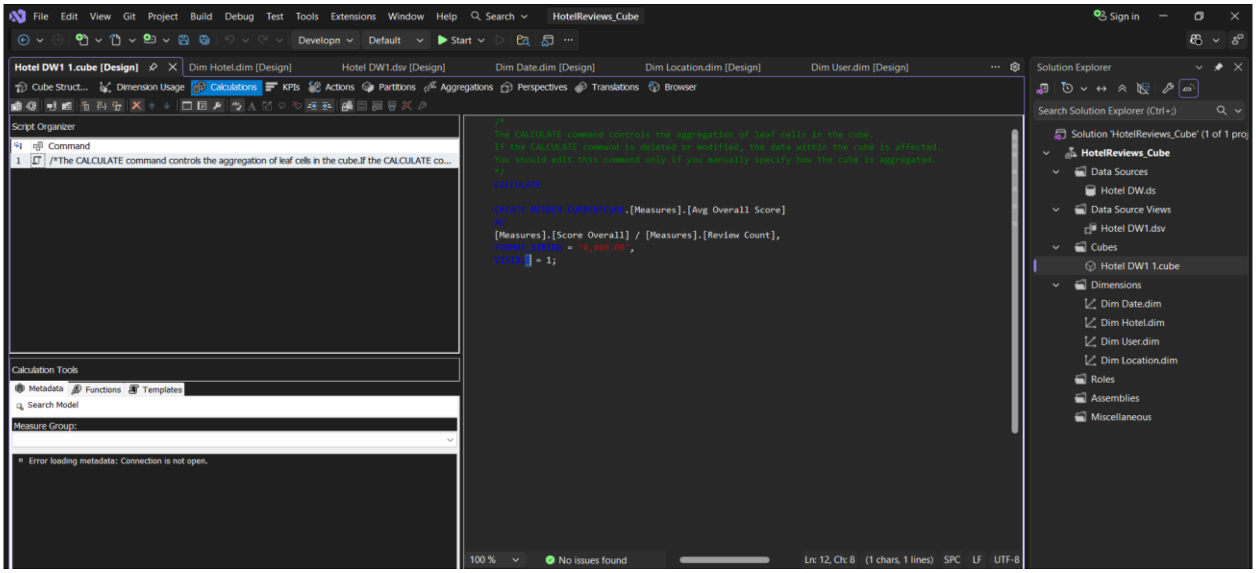
2.4 Creating the Cube

The Cube Wizard was used to create the OLAP cube using the Fact_Reviews table as the measure group. Key measures such as review scores were selected for analysis, and review_id was configured as a count measure. All dimensions were successfully linked to the cube, enabling multidimensional analysis.



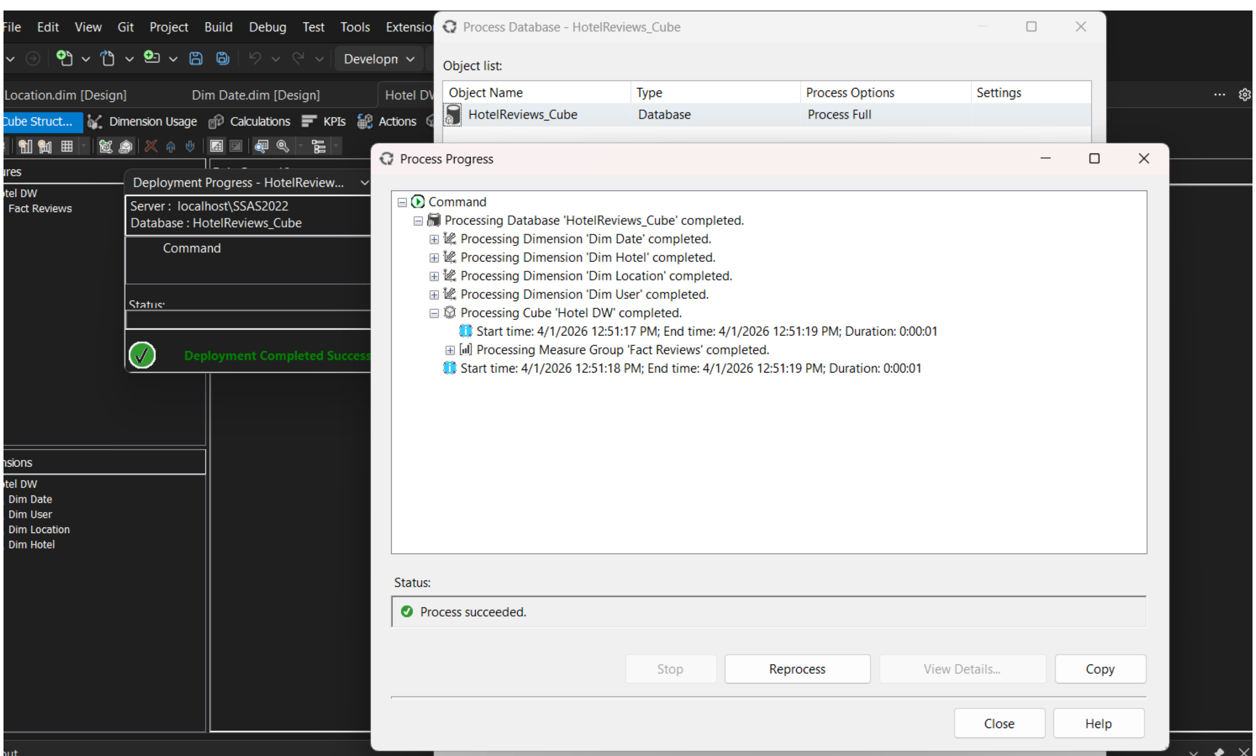
2.5 Calculated Measure

A calculated measure named “Avg Overall Score” was created using MDX in the Calculations tab. This measure divides the total Score Overall by the Review Count to calculate the average rating. The result was formatted to two decimal places for better readability in reports.



2.6 Deploy and Process Cube

The cube was deployed to the Analysis Services server and processed using the “Process Full” option. This step loads data from the data warehouse into the cube and prepares it for analysis. Successful processing ensures that the cube is ready for querying



The successful deployment and validation of the cube confirm that the data warehouse supports efficient OLAP operations, enabling users to perform time-based, location-based, and user-based analysis of hotel reviews.

Cube Summary

Component	**Details**
Cube Name	HotelReviews_Cube
Data Source	Hotel_DW (SQL Server)
Fact Table	Fact_Reviews — 49,999 rows
Measures (7)	score_overall, score_cleanliness, score_comfort, score_facilities, score_location, score_staff, score_value_for_money
Calculated Measure	Avg Overall Score = Score Overall / Review Count
Count Measure	Review Count (from review_id)
Dimension 1	Dim_Date — Hierarchy: Year → Month → Day
Dimension 2	Dim_Hotel — Hierarchy: Country → City → Hotel Name
Dimension 3	Dim_User — Attributes: gender, country, age_group, traveller_type
Dimension 4	Dim_Location — Hierarchy: Continent → Country → City

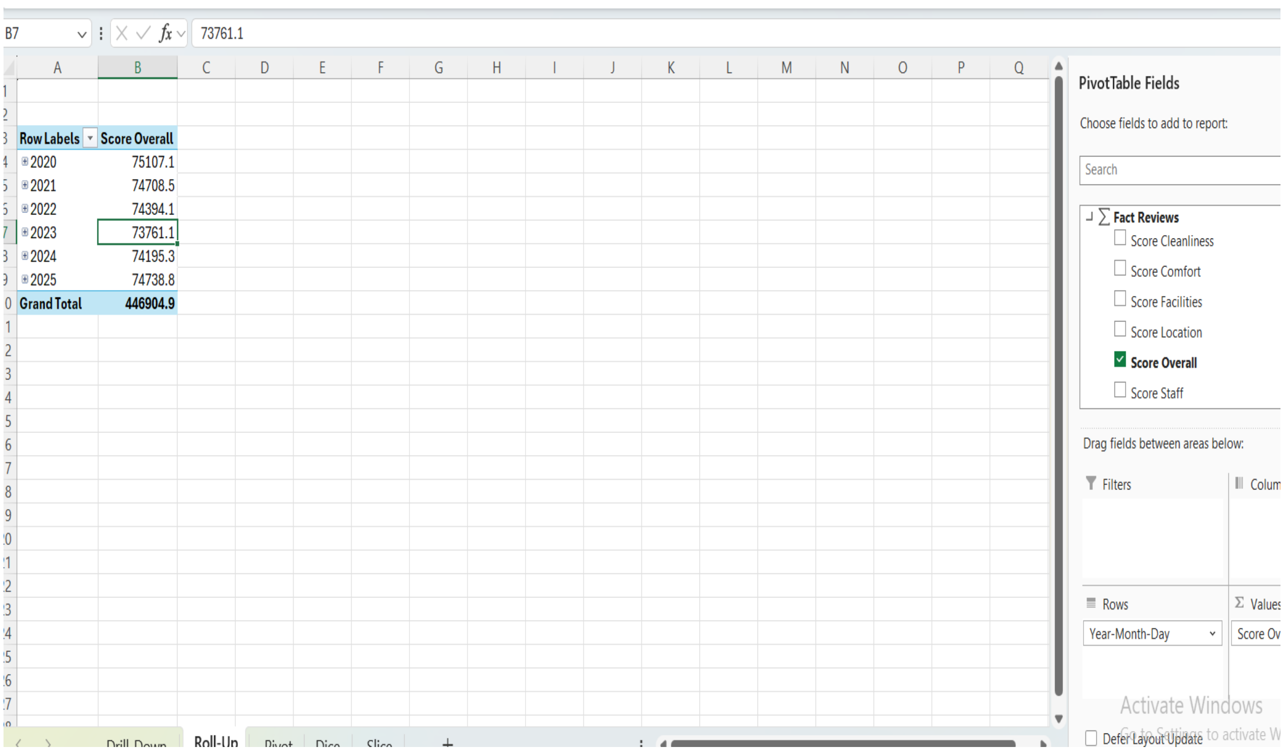
3 Demonstration of OLAP operations

Microsoft Excel was connected to the SSAS cube using the Analysis Services connection. The HotelReviews_Cube database and cube were selected successfully, enabling Excel to retrieve multidimensional data for analysis.

A PivotTable was created in a new worksheet, displaying all available dimensions and measures from the cube. This interface allows users to drag and analyze data dynamically across different dimensions.

3.1 OLAP Operation 1 — Roll-Up

The roll-up operation aggregates data to a higher level in the hierarchy. By placing Year in rows and Review Count in values, monthly data is summarized into yearly totals, providing a high-level overview of review trends.

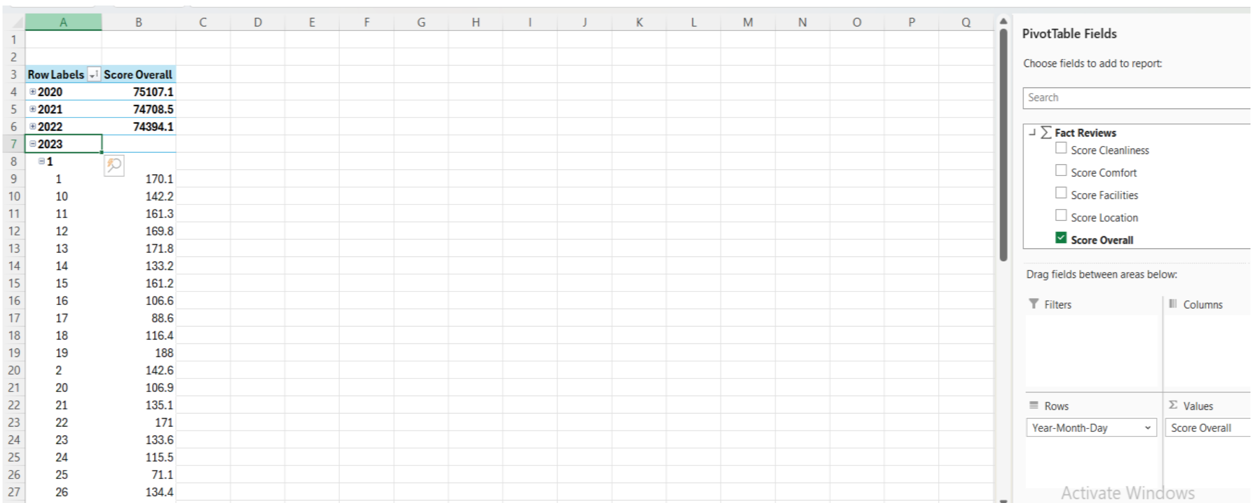


3.2 OLAP Operation 2 — Drill-Down

The initial view shows aggregated review counts at the Year level.

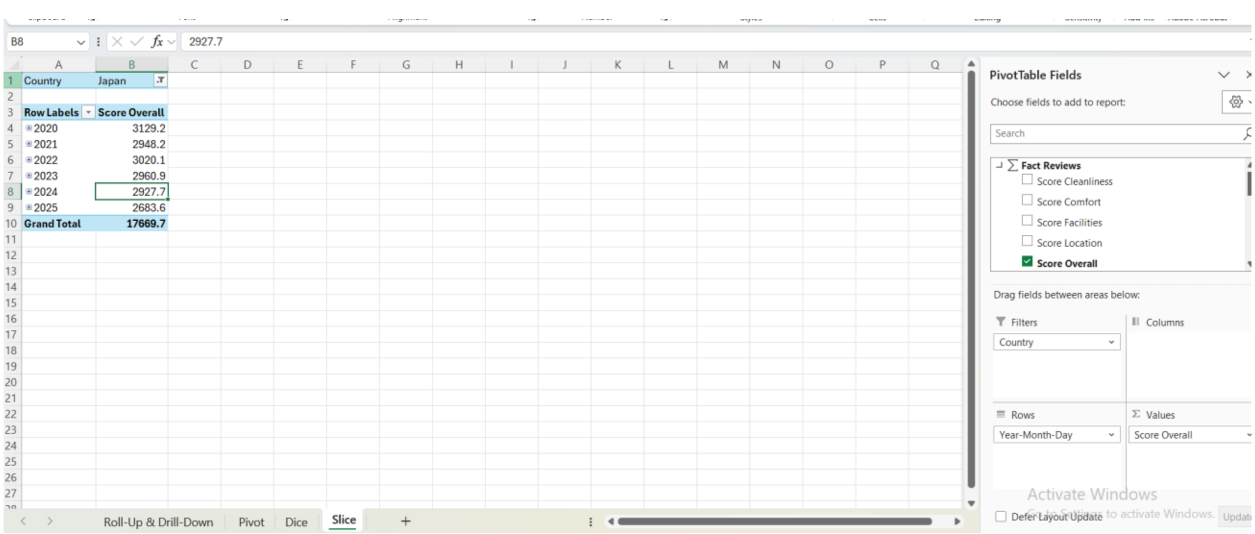
Drilling down into a selected year expands the data to display monthly review counts, providing more detailed insights.

Further drilling down reveals daily-level data, allowing detailed analysis of review patterns over time.



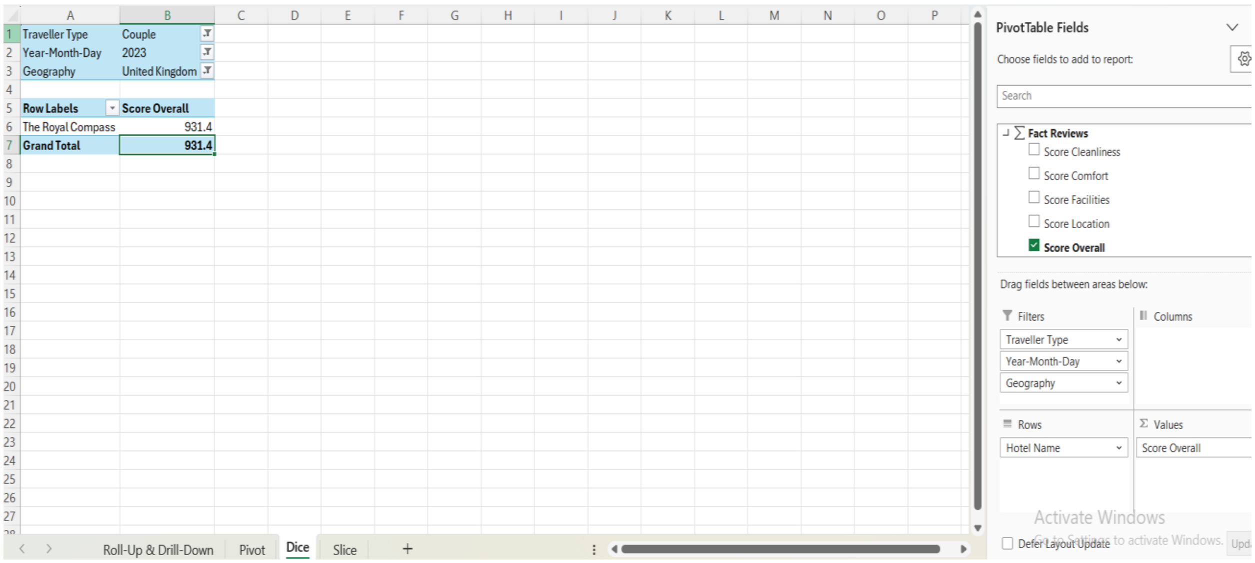
3.3 OLAP Operation 3 — Slice

The slice operation filters the cube using a single dimension. By selecting Country = Japan, the PivotTable displays only review data related to Japanese hotels, enabling focused analysis on a specific subset.



3.4 OLAP Operation 4 — Dice

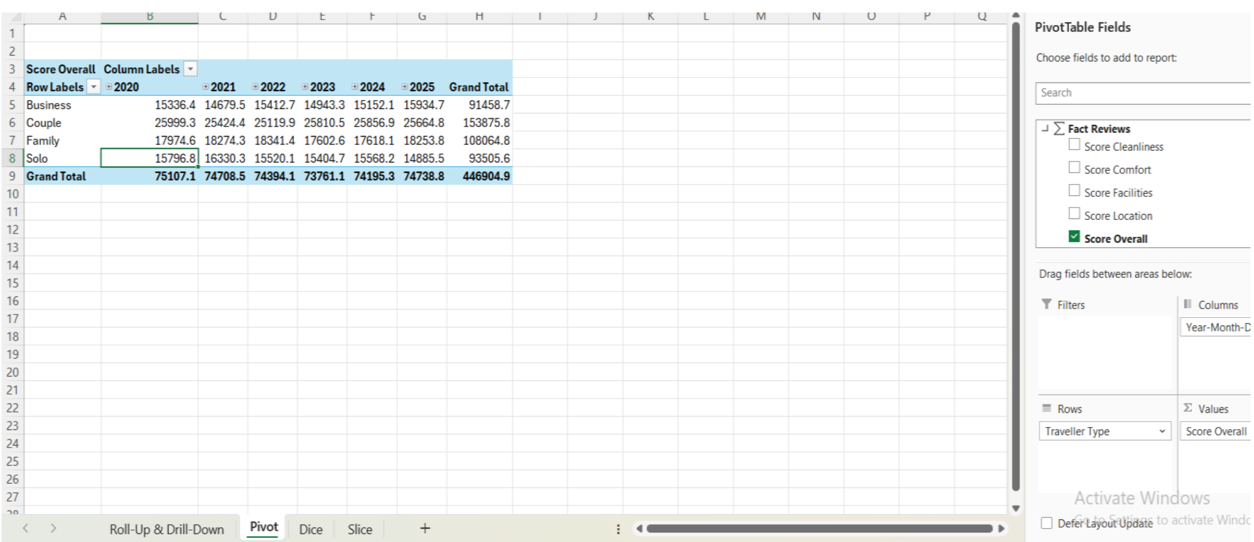
The dice operation applies multiple filters simultaneously to create a sub-cube. In this case, filters were applied for Country = United States, Traveller Type = Couple, and Year = 2023, allowing highly specific analysis of selected data.



3.5 OLAP Operation 5 — Pivot

In the initial view, Year is placed in rows and Traveller Type in columns, showing average scores across both dimensions.

The pivot operation rotates the axes by swapping rows and columns. This provides a different perspective of the same data without changing the underlying values.

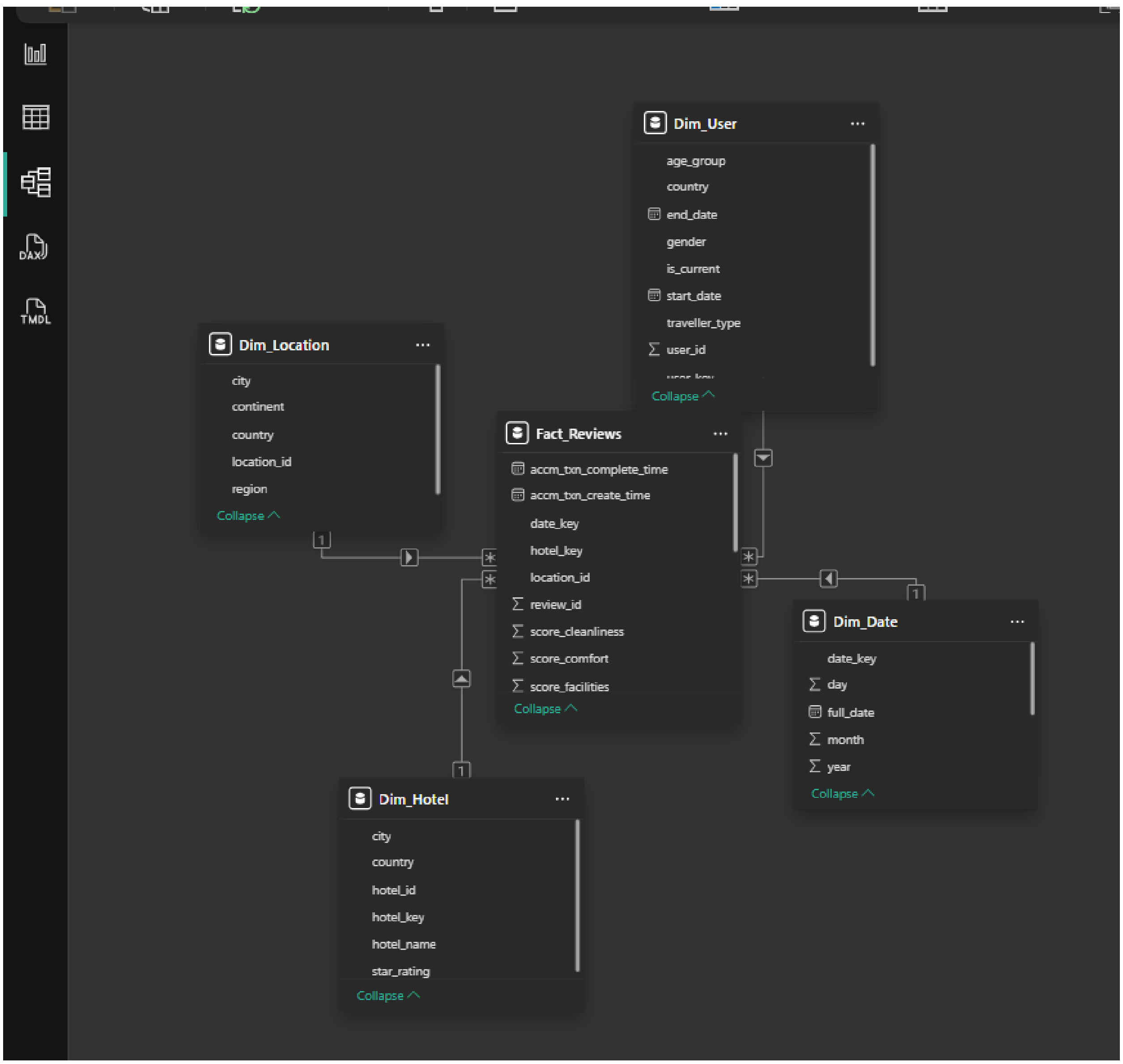


4 PowerBI Reports

4.1 Data Preparation

The Hotel_DW SQL Server database was connected to Power BI Desktop using Get Data → SQL Server with server name DESKTOP-C9S2LPN\ASUS and database Hotel_DW in Import mode. Four tables were loaded: Fact_Reviews, Dim_Hotel, Dim_User, Dim_Location and Dim_Date. In Power Query Editor, the Dim_User table was filtered to keep only active users where is_current = 1 before clicking Close & Apply.

4.2 Data Modeling

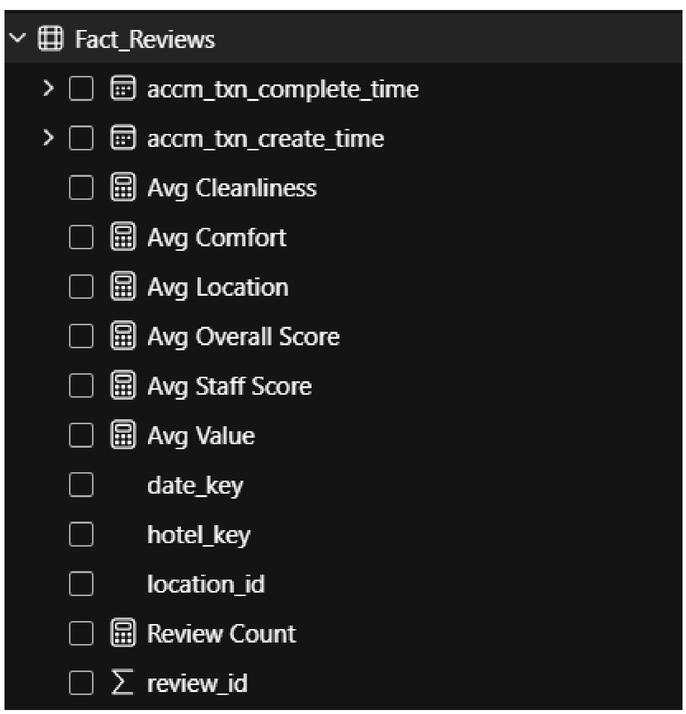


The Model View confirmed the star schema relationships:

- Fact_Reviews[user_key] → Dim_User[user_key]
- Fact_Reviews[hotel_key] → Dim_Hotel[hotel_key]
- Fact_Reviews[date_key] → Dim_Date[date_key]
- Fact_Reviews[location_id] → Dim_Location[location_id]

4.3 DAX Measures

Measure Name	DAX Formula
Total Reviews	Total Reviews = COUNTROWS(Fact_Reviews)
Avg Overall Score	Avg Overall Score = AVERAGE(Fact_Reviews[score_overall])
Avg Cleanliness	Avg Cleanliness = AVERAGE(Fact_Reviews[score_cleanliness])
Avg Comfort	Avg Comfort = AVERAGE(Fact_Reviews[score_comfort])
Avg Staff Score	Avg Staff Score = AVERAGE(Fact_Reviews[score_staff])
Avg Value Score	Avg Value = AVERAGE(Fact_Reviews[score_value_for_money])
Avg Location Score	Avg Location = AVERAGE(Fact_Reviews[score_location])



The presentation layer of the analytical solution was developed using Power BI Service and Power BI Desktop. The reports connect directly to the underlying SSAS Multidimensional Cube via a Live Connection, ensuring that all data modeling, hierarchies, and security rules established in the data warehouse are preserved and instantly accessible. The documentation below details the data preparation, modeling strategies, and visual design for each required report.

4.4 Report 1 — Matrix Visual

Purpose: This report is to display detailed tabular data using a Matrix visual with hierarchical row and column groupings. This enables comparison of hotel performance across multiple years while analyzing score breakdowns.

Steps followed: A new report page was created and renamed “Report 1 – Matrix”. A Matrix visual was inserted and configured with Dim_Hotel[country] and Dim_Hotel[hotel_name] in the Rows section, creating a hierarchical structure where countries can be expanded to show individual hotels.

Dim_Date[year] was added to the Columns section to display yearly data from 2020 to 2025. The Values section included Review Count, Avg Overall Score, Avg Staff Score, and Avg Cleanliness, allowing multiple performance metrics to be analyzed simultaneously.

Formatting applied: Row and column subtotals were enabled to display aggregated totals at both country level and overall level. Conditional formatting was applied to Avg Overall Score using a green color gradient, where darker shades represent higher scores. The visual was titled “Hotel Score Performance by Country and Year” to clearly indicate its purpose.

What it shows: The matrix visual allows users to expand each country to view individual hotel performance, providing a clear hierarchical view. The year-based columns support cross-year comparison of review metrics. The conditional formatting highlights high-performing hotels, making it easy to identify trends and top-rated locations at a glance.

Hotel Score Performance by Country and Year

year	2020				2021				2022
hotel_name	Review Count	Avg Overall Score	Avg Staff Score	Avg Cleanliness	Review Count	Avg Overall Score	Avg Staff Score	Avg Cleanliness	Review
	349	**8.76**	**8.77**	**8.76**	**317**	**8.75**	**8.76**	**8.78**	
Mexico	349	8.76	8.77	8.76	317	8.75	8.76	8.78	
	338	**8.99**	**9.01**	**9.19**	**334**	**9.00**	**9.03**	**9.18**	
Germany	338	8.99	9.01	9.19	334	9.00	9.03	9.18	
	337	**9.09**	**9.10**	**9.23**	**303**	**9.09**	**9.10**	**9.26**	
Netherlands	337	9.09	9.10	9.23	303	9.09	9.10	9.26	
	308	**9.04**	**8.97**	**9.15**	**345**	**9.04**	**8.94**	**9.14**	
Italy	308	9.04	8.97	9.15	345	9.04	8.94	9.14	
	323	**8.88**	**8.83**	**9.01**	**326**	**8.87**	**8.86**	**8.95**	
Brazil	323	8.88	8.83	9.01	326	8.87	8.86	8.95	
	327	**9.02**	**8.90**	**9.14**	**334**	**9.04**	**8.96**	**9.15**	
Spain	327	9.02	8.90	9.14	334	9.04	8.96	9.15	
	346	**9.02**	**9.20**	**9.23**	**340**	**9.02**	**9.16**	**9.24**	
South Korea	346	9.02	9.20	9.23	340	9.02	9.16	9.24	
	327	**8.96**	**9.03**	**9.04**	**311**	**8.97**	**9.03**	**9.07**	
Russia	327	8.96	9.03	9.04	311	8.97	9.03	9.07	

Page 1 | Report 1 - Matrix | Report 2 - Slicers | Report 3 - Drill Down | Report 4 - Summary | Hotel Detail Page

Activate Windows
Go to Settings to

Matrix report

4.5 Report 2 — Cascading Slicers

Purpose: The purpose of this task is to demonstrate cascading filtering using slicers and to visualize data dynamically through multiple chart types. This allows users to interactively explore the dataset and observe how filtering one dimension affects other related data.

Steps followed: A new report page was created and renamed “Report 2 – Slicers”. Three slicers were added to enable interactive filtering. The first slicer was created using Dim_Hotel[country] and formatted as a dropdown, titled “Select Country”. The second slicer was created using Dim_Hotel[hotel_name], titled “Select Hotel”. Since both fields belong to the same dimension table, the hotel slicer automatically cascades based on the selected country.

The third slicer was created using Dim_Date[year] and configured in “Between” mode, allowing users to select a range of years using a slider. This enables time-based filtering of the data.

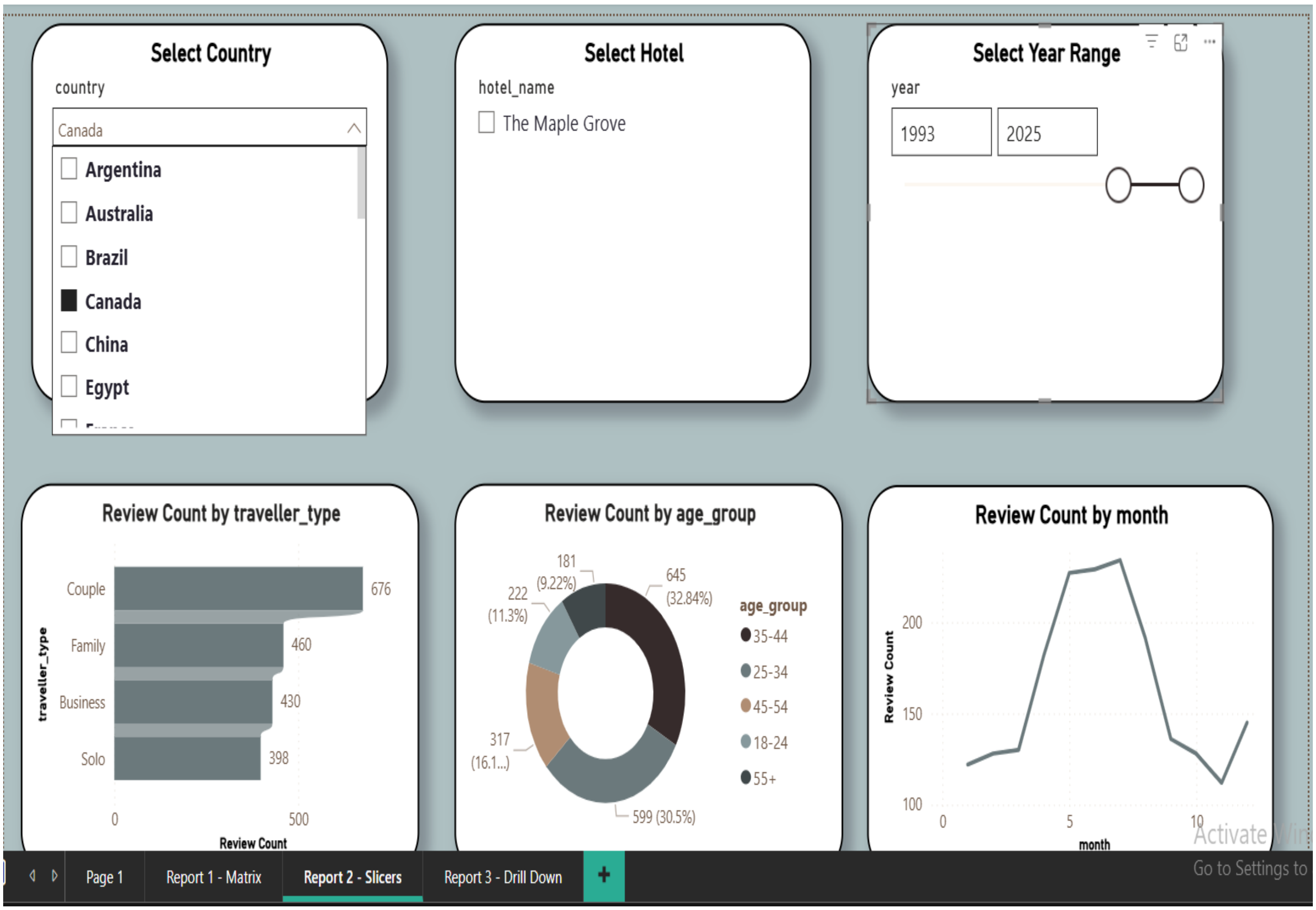
Three chart visuals were added:

- A Clustered Bar Chart displaying Review Count by traveller_type to compare review distribution across different traveler categories.
- A Donut Chart showing Review Count by age_group to illustrate the proportion of reviews across age groups.
- A Line Chart presenting Review Count by month to analyze trends and patterns over time.

Cascading filter test:

The cascading filter functionality was verified by selecting a country in the first slicer. For example, selecting “Canada” automatically filters the hotel slicer to display only hotels located in Canada, such as “The Maple Grove”. This confirms that slicers are interconnected and respond dynamically based on shared dimensions.

Additionally, all chart visuals update instantly according to the selected filters, ensuring real-time interaction and consistent data representation across the report.



Slicers report

4.6 Report 3 — Drill-Down Hierarchy Report

Purpose: The purpose of this report is to enable users to explore review data hierarchically using drill-down functionality. This allows navigation from high-level summaries (Year) to more detailed levels (Month and Day), supporting in-depth time-based analysis**.**

Steps followed: A Date Hierarchy was created in the Data pane by right-clicking the year field and selecting “Create Hierarchy”, then adding month and day to form a Year → Month → Day structure. A new report page was created and renamed “Report 3 – Drill Down”.

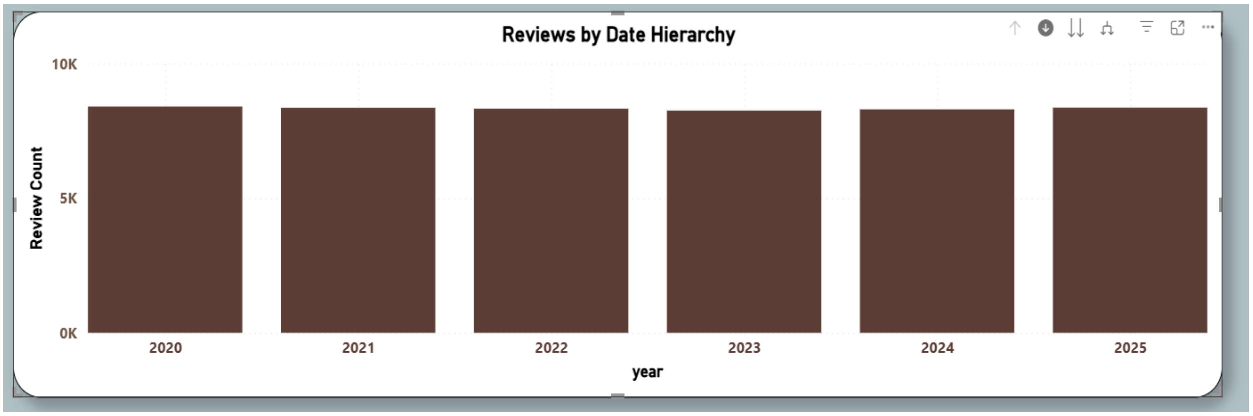
Chart 1 - Date Hierarchy Drill Down: A Clustered Column Chart was created using the Date Hierarchy on the X-axis and Review Count on the Y-axis. The built-in drill-down controls allow users to navigate through hierarchy levels. Clicking the double down arrow (↓↓) expands all data to the next level (e.g., Year → Month), while clicking the single down arrow (↓) enables drilling into a specific period.

Chart 2 - A second Clustered Column Chart was created using Dim_Hotel[country] on the X-axis and Avg Overall Score on the Y-axis. This visual provides a comparative view of hotel performance across multiple countries, supporting location-based analysis**.**

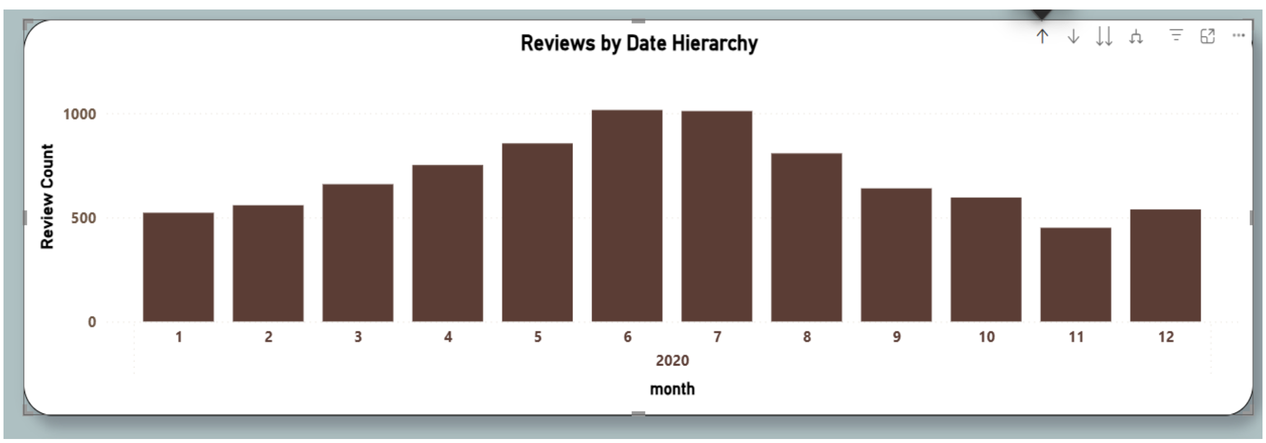
What it shows: The report allows users to start from a high-level yearly view (2020–2025) and progressively drill down into monthly and daily review data. This helps identify trends, seasonal patterns, and detailed variations in review activity. The geographic chart complements this by showing how performance differs across countries.

Drill Down report

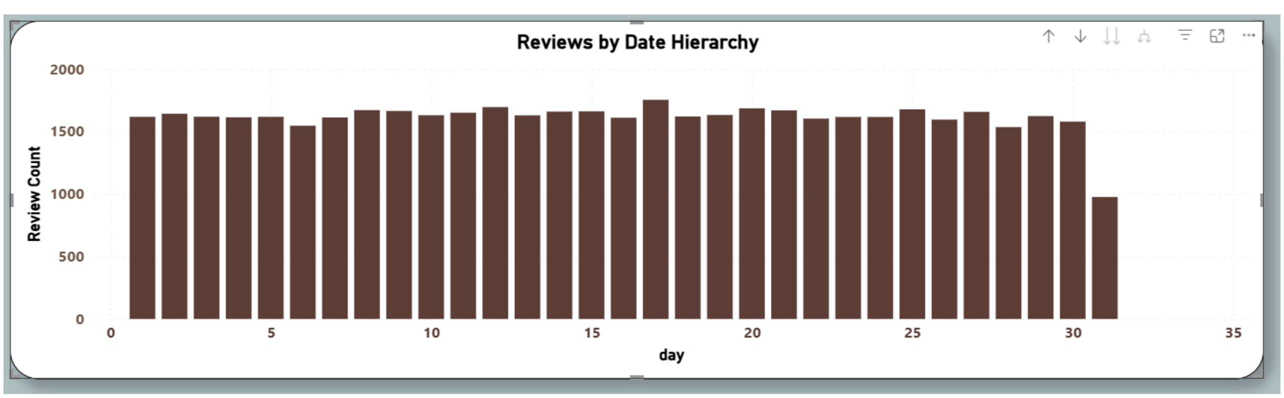
This visualization shows the total review count aggregated at the Year level (2020–2025), providing a high-level overview of review trends over time.



Drilling down into a selected year reveals monthly review counts, allowing identification of seasonal patterns and variations within the year.



Further drill-down displays daily review counts, enabling detailed analysis of fluctuations and peak activity periods at the most granular level.



4.7 Report 4 — Drill-Through Report

Purpose: The purpose of this report is to demonstrate drill-through functionality, allowing users to navigate from a summary view to a detailed report for a selected hotel. This enables focused analysis of individual hotel performance.

Steps followed: A summary page was created and renamed "Report 4 – Summary". A Table visual was added containing country, hotel_name, Review Count, Avg Overall Score, and Avg Staff Score. The table was sorted in descending order based on Avg Overall Score to highlight top-performing hotels. The title was set to "Hotel Summary — Click a Hotel to Drill Through".

A second page named "Hotel Detail Page" was created to display detailed information. With the canvas selected, Dim_Hotel[hotel_name] was added to the Drill-through field, enabling drill-through functionality. A Back button was automatically generated to support navigation back to the summary page.

Summary Page: A new page was created and renamed **Report 4 - Summary**. A Table visual was added with columns: country, hotel_name, Review Count, Avg Overall Score and Avg Staff Score, sorted by Avg Overall Score descending. Title set to **"Hotel Summary — Click a Hotel to Drill Through"**.

Hotel Summary — Click a Hotel to Drill Through

country	hotel_name	Review Count	Avg Overall Score	Avg Staff Score
United Arab Emirates	The Golden Oasis	2035	9.09	9.24
Netherlands	Canal House Grand	1948	9.09	9.10
Singapore	Marina Bay Zenith	2013	9.05	9.04
China	The Bund Palace	2038	9.05	8.97
Italy	Colosseum Gardens	1971	9.04	8.97
Spain	Gaudi's Retreat	2038	9.04	8.94
Canada	The Maple Grove	1964	9.04	9.10
South Korea	Han River Oasis	2069	9.02	9.17
New Zealand	The Kiwi Grand	2059	9.02	9.02
Thailand	The Orchid Palace	2052	9.01	9.04
Germany	Berlin Mitte Elite	2021	9.00	9.03
South Africa	Table Mountain View	1978	8.98	9.10
Japan	Kyo-to Grand	1970	8.97	9.32
Russia	Kremlin Suites	1969	8.97	9.03
France	L'+ëtoile Palace	1991	8.95	9.15
Turkey	The Bosphorus Inn	1948	8.91	8.89
Australia	Sydney Harbour Grand	1954	8.90	8.89
United Kingdom	The Royal Compass	1900	8.88	8.95
Brazil	Copacabana Lux	1960	8.87	8.82
India	The Gateway Royale	1986	8.84	8.82
United States	The Azure Tower	1997	8.84	8.69
Argentina	Tango Boutique	1955	8.82	8.82
Total		**49970**	**8.94**	**8.97**

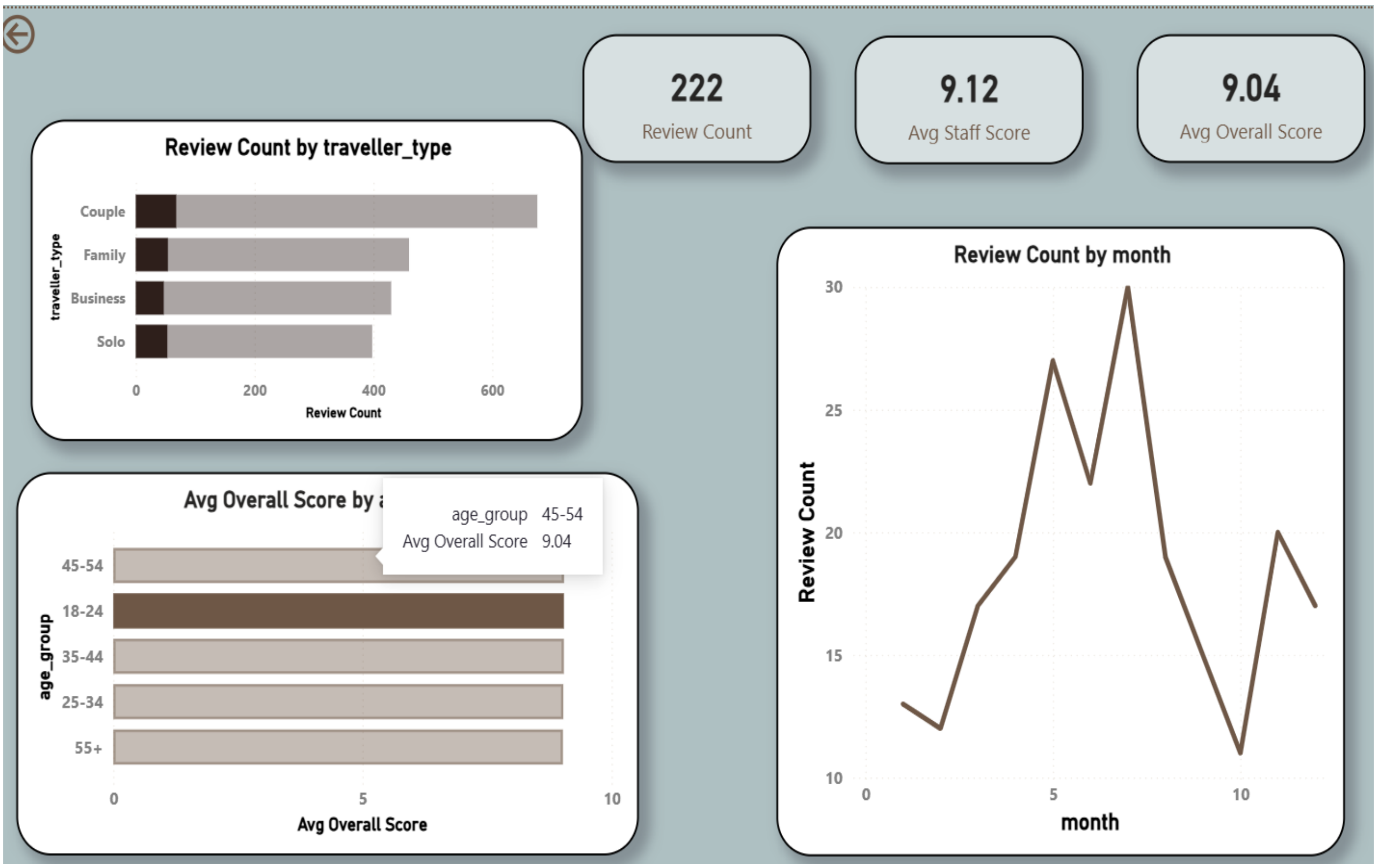
- Matrix | Report 2 - Slicers | Report 3 - Drill Down | Report 4 - Summary | Hotel Detail Page

Summary page

Hotel Detail Page: A new page was created and renamed **Hotel Detail Page**. With the canvas selected (no visual), Dim_Hotel[hotel_name] was dragged into the **Drill through** field box in the Visualizations pane — this automatically created a Back button in the top-left corner.

Six visuals were added to the detail page:

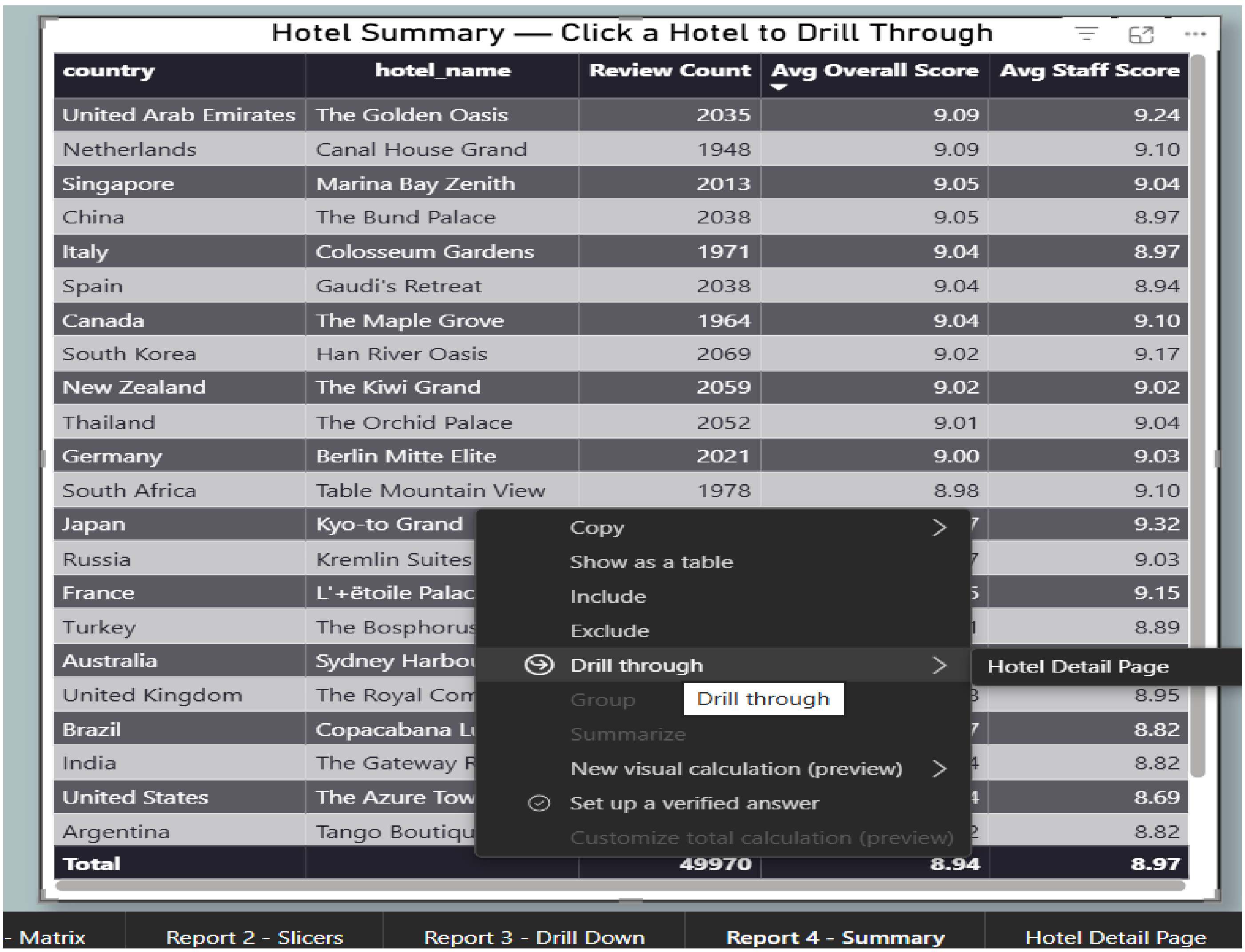
- Card visuals displaying Review Count, Avg Overall Score, and Avg Staff Score
- A Line Chart showing monthly review trends
- A Bar Chart showing Review Count by traveller type
- A Bar Chart showing Avg Overall Score by age group



Detail Page

Testing: Right-clicking any hotel name in Report 4 - Summary reveals the context menu **Drill through → Hotel Detail Page**. After drilling through, all visuals update to show data exclusively for the selected hotel. The Back button returns the user to the summary page.

Drill-Through Functionality: The drill-through feature was tested by right-clicking a hotel name in the summary table and selecting Drill through → Hotel Detail Page. The detail page dynamically filtered all visuals to display data for the selected hotel only. The Back button allows users to return to the summary page seamlessly.



What it Shows: This report enables users to transition from a high-level overview to a detailed analysis of a specific hotel. It supports better decision-making by allowing users to investigate performance trends, customer segments, and time-based patterns at an individual hotel level.

Conclusion

In this assignment, a data warehouse and OLAP solution was successfully developed for hotel review analysis. A star schema was designed with fact and dimension tables to organize the data effectively.

The SSAS cube was created to support multidimensional analysis, and operations such as roll-up, drill-down, slice, dice, and pivot were performed using Excel. These operations helped to analyze the data from different perspectives.

Power BI was used to create interactive reports including matrix, drill-down, slicers, and drill-through reports. These visualizations made it easier to explore and understand the data.

Overall, this assignment demonstrates how data warehousing and business intelligence tools can be used to convert raw data into meaningful insights for analysis and decision-making.